

Business Need	Proposed Response	Expected Outcome
Fragmented station records	One centralized cloud platform with role and station scoping	Real-time visibility and consistent records across locations
Inventory and transaction errors	Controlled stock movements, approvals, audit logs and reconciliation	Improved stock accuracy and reduced manual correction
Limited financial visibility	Agent wallets, income/expense tracking, cashbook, profit and station revenue reports	Clearer profitability and financial accountability
Manual cargo documentation	AWB, barcode, QR code and printer-ready cargo labels	Faster processing and traceable cargo records
Weak operational oversight	Executive, business-unit, station and role-based dashboards	Data-driven decisions and measurable accountability
Security and accountability gaps	RBAC, authentication controls, immutable audit trail and encrypted storage	Stronger governance, traceability and reduced misuse

2. Project Objectives

Objective	Success Definition
Centralize operations	All approved stations use one authoritative platform for inventory, sales, cargo, customers, staff, agents and finance.
Strengthen control	Role permissions, approvals, audit trails and correction workflows are applied to sensitive actions.
Improve speed	Barcode scanning, reusable customer records, automated labels, receipts and reports reduce processing time.
Improve transparency	Management receives current sales, stock, station, agent, staff and profitability information.
Support growth	The architecture supports new stations, business units, airlines, payment methods and future integrations.
Protect information	Sensitive records and files are protected through secure authentication, encryption, signed file access and backups.

Final Functional Scope: The following table defines the agreed functional modules. Detailed workflows, field labels, validations and permission boundaries will be finalized during Week 1 discovery and UI/UX approval without reducing the listed capabilities.

No.	Module	Included Functionality
1	Executive & Operational Dashboards	Real-time sales, inventory value, low/out-of-stock alerts, customers, agents, staff, stations, pending transactions, recent activities, revenue graphs and sales trends. Separate dashboards for Partners like Binani, UMZA, Logistics and future Clearing & Forwarding, plus consolidated management reporting.
2	Station Administration	Create, edit, disable and configure stations; assign station managers; view performance; scope users and transactions; transfer inventory between stations; add future stations without structural redevelopment.
3	User Access & RBAC	User creation, activation/deactivation, assigned station, username/email authentication, password reset, role assignment and granular permissions for Super Admin, Admin, Finance, HR, Operations Manager, Sales Officer, Operations Coordinator, Customer Service and Auditor.
4	Staff & HR Records	Staff registration, staff ID, department, position, contact details, salary, employment date, status, station assignment, passport photograph and optional next-of-kin records.

		Sensitive HR fields restricted by permission.
5	Customer Management	Internal customer registration before sales; name, phone, optional email, PNR, optional national ID, destination, airline and remarks; returning-customer search, transaction history and duplicate-management tools.
6	Purchase & Inventory	Product catalogue, product code, barcode/QR, category, supplier, purchase/selling price, quantity, unit, minimum/reorder levels, expiry, batch, station and purchase date. Stock-in/out, transfer, adjustment, damage, return and full movement history.
7	Point of Sale	Product search and scanning, customer and agent selection, permission-controlled discounts, multi-method payment, transaction numbering, invoice and receipt generation, refunds, cancellations and outstanding-payment tracking.
8	Cargo & AWB Labeling	AWB/cargo number, sender/customer and receiver details, origin, destination, weight, pieces, commodity, airline, flight number, date and handling notes. Searchable, reprintable thermal/A5 labels with barcode and QR code.
9	Sales & Revenue Management	Daily, weekly, monthly and yearly sales views; filters by station, airline, sales officer, customer, payment method and business unit; total sales, refunds, cancellations and outstanding payments; PDF, Excel and CSV exports.
10	Agent Management & Wallets	Agent profile, company and contact details, address, credit limit, opening balance, deposits, wallet credit/debit, transaction history, outstanding balance, statements, print/export and finance/admin oversight.
11	Financial Management	Daily income and expenses, cashbook, agent deposits, wallet transactions, refund management, profit reporting, sales summaries, station revenue and finance dashboard. Controlled reversals and approval workflows.
12	Reporting & Analytics	Reports by station, user, product, agent, customer, airline, date and business unit; dashboard KPIs; filters, print views, PDF/Excel/CSV export and permission-aware report access.
13	Printing & Documents	Cargo labels, sales receipts, invoices, agent statements, sales reports and inventory reports optimized for thermal and A5 printers. Electronic copies stored against transactions where applicable.
14	Notifications	Automatic alerts for low stock, low agent balance, large transactions, user creation, failed login attempts, pending approvals and inventory adjustments. In-app notification foundation with email/SMS integration readiness.
15	Audit Trail	Immutable event logging for logins, product changes, sales, refunds, adjustments, price changes, user changes, approvals and management corrections, including actor, date/time, station and IP address where available.
16	Management & Correction Tools	Authorized correction of transactions and inventory, sales reversal, label edits, password reset, account locking, stock transfer, duplicate customer merge, inventory recalculation, diagnostics, backup/restore references and data-repair workflows. Every correction is audited.
17	Security Controls	Role-based authorization, password hashing, optional two-factor authentication readiness, session timeout, data encryption, permission control, activity logging, secure file access and automated backup procedures.
18	Global Search	Permission-aware search across customers, agents, products, receipts, labels, staff, stations, users, transactions, PNR and AWB number.
19	System Configuration	Company identity and logo, receipt footer, tax settings, payment methods, business units, stations, permissions, printer settings and email/SMS notification settings.
20	Advanced & Future-Ready Features	Barcode scanner support, real-time multi-user synchronization, offline-capable PWA workflow with queued synchronization, customer history, agent performance, inventory forecasting, product movement analysis, purchase recommendations, daily backups, document storage and future accounting, SMS, email and airline APIs.

21	Flight Ticket Booking	Internal booking records for customers, PNR, airline, route, travel date, passenger/contact details, fare, service charge, payment status, booking status, ticket/itinerary uploads, receipt generation, profit tracking and future airline/GDS API readiness.
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4. Roles and Access Model

Role	Typical Scope	Key Access
Super Admin	All stations and modules	Full configuration, users, permissions, corrections, audit and consolidated reporting.
Admin	Company-wide or delegated scope	Operations oversight, station management, reports, users and approvals as assigned.
Finance	Assigned stations or company-wide	Payments, cashbook, deposits, expenses, wallets, refunds, profit and financial reports.
HR	Staff and employment records	Staff registration, updates, documents and employment status; no unrestricted finance access.
Operations Manager	Cargo, inventory and station operations	Inventory, transfers, cargo records, labels, approvals and operational reports.
Sales Officer	Assigned station	Customer registration, POS sales, authorized payments and receipt generation.
Operations Coordinator	Assigned operational territory	Coordinate cargo and station workflows, monitor exceptions and performance.
Customer Service	Customer and transaction enquiry	Read-focused access to customers, transactions, PNR, cargo and approved updates.
Auditor	Read-only oversight	Reports, audit logs and historical records without transaction mutation.
Custom Role	As configured	Granular permissions assembled by authorized administrators.

Access Control Principle

Permissions will be enforced on the server and database query scope, not only by hiding menu items. Station-level users will see only assigned station data unless a broader scope is explicitly granted.

5. Proposed Technical Architecture

Layer	Technology / Service	Implementation Notes
Application Framework	Next.js (App Router) + TypeScript	Full-stack web application using server components, server actions and route handlers where appropriate.
User Interface	Tailwind CSS + shadcn/ui	Responsive enterprise interface with reusable components, accessible forms, data tables, light/dark mode and print layouts.
Authentication	Auth.js	Secure credentials flow, session handling, password reset, account status, role and permission checks.
Database	PostgreSQL (managed cloud)	Production relational database; recommended deployment on Neon PostgreSQL or an approved equivalent.
ORM & Migrations	Prisma ORM	Typed data access, schema migrations, transaction handling and seed scripts.
File Storage	Amazon S3 bucket	Private object storage for staff photos, IDs, receipts, cargo labels, ticket documents and attachments using signed

		upload/download URLs.
Application Hosting	Vercel	Production, staging and preview deployments with HTTPS, environment separation, logs and controlled release workflow.
Background Processing	Scheduled/queued job abstraction	Automated notifications, backups, reporting jobs, synchronization processing and heavy exports.
Documents	Server-generated PDF/print templates	Receipts, invoices, labels, agent statements and reports with QR/barcode support.
Monitoring	Vercel logs + optional Sentry	Application errors, deployment health, performance monitoring and diagnostic tracing.
Email/SMS	Provider adapters	Architecture ready for SMTP/Resend and Nigerian SMS providers; provider fees are separate.
Future Integrations	REST/Webhook adapter layer	Prepared for accounting, airline/GDS, payment, SMS and external business systems.

5.1 Recommended Data Flow

Step	Flow
1	Authorized user accesses the secured Next.js application through HTTPS on the company domain.
2	Auth.js validates the user session, account status, role, permission and station scope.
3	Validated operations execute through server-side services and Prisma transactions against PostgreSQL.
4	Files are uploaded directly to a private AWS S3 bucket through short-lived signed URLs; only file metadata is stored in the database.
5	Reports, receipts and labels are generated from controlled templates and linked to the originating record.
6	Audit events are written for sensitive actions, including corrections, approvals, refunds and access events.
7	Vercel deploys validated releases from the source repository to preview, staging and production environments.

Deployment and Infrastructure Specification: The production environment will use Vercel for the Next.js application and Amazon S3 for private file storage. The database will use a managed PostgreSQL service. This separation provides reliable application delivery, scalable storage and simpler operational maintenance.

Environment	Purpose	Configuration
Development	Local engineering and feature development	Local environment variables, development database, test S3 prefix/bucket and seeded test data.
Preview	Automatic per-branch validation	Vercel preview deployment for each approved code branch or pull request; never connected to production data.
Staging / UAT	Client testing and acceptance	Dedicated Vercel project/environment, staging database, staging S3 prefix/bucket and UAT user accounts.
Production	Live company operations	Custom domain, production PostgreSQL, private production S3 bucket, production secrets, monitoring, backups and controlled deployments.

6.1 Vercel Deployment

Area	Final Specification
Hosting	Vercel Pro is recommended for a business-critical production deployment, subject to current provider pricing and limits.

Domain	Client-provided domain or subdomain, such as operations.aauchamo.com, connected to Vercel with automatic TLS/HTTPS.
Release Process	Git-based deployment with preview review, staging validation, database migration check and controlled promotion to production.
Environment Variables	Separate encrypted values for development, preview, staging and production. Secrets will not be committed to source control.
Server Runtime	Node.js runtime for database, authentication, PDF generation and S3 signing operations; edge runtime used only where suitable.
Logs & Diagnostics	Vercel runtime and deployment logs, health checks and optional Sentry integration.
Scaling	Automatic application scaling within Vercel plan limits; database and S3 scale independently.
Rollback	Previous stable Vercel deployment can be restored rapidly; database changes follow migration and backup procedures.

6.2 AWS S3 File Storage

Control	Implementation
Bucket Access	Private bucket; public listing and anonymous access disabled.
Upload Method	Short-lived presigned URLs to avoid passing large files through the application server.
Access Method	Permission-checked signed download URLs with short expiration.
Organization	Structured object keys by environment, module, station, record type, year and record ID.
Encryption	S3 server-side encryption enabled. AWS KMS may be used where enhanced key control is required.
Versioning	Recommended for recovery from accidental overwrite or deletion.
Lifecycle	Configurable transition or deletion rules for obsolete temporary files and long-term archives.
Validation	File type, extension, size and ownership validation before metadata is accepted by the application.
Auditability	Uploader, record association, file hash/metadata, creation date and access permission stored in the database.
Backup Position	S3 durability does not replace governance; versioning, lifecycle and optional cross-region replication will be agreed based on budget.

6.3 Database and Backup

Area	Specification
Database	Managed PostgreSQL with encryption in transit and at rest, connection pooling and environment isolation.
Data Model	Normalized schema for stations, users, staff, customers, products, stock movements, sales, cargo, agents, wallets, finance, tickets, reports and audit events.
Backups	Daily automated provider backups plus agreed periodic export/restore testing. Retention period to be confirmed before production.
Migration	Version-controlled Prisma migrations executed first in staging and then production through a controlled release.
Recovery	Documented recovery procedure for application rollback, database restore and file recovery.
Retention	Financial, audit and transaction records will be retained according to client policy and applicable obligations; hard deletion will be restricted.

7. Security, Privacy and Audit Controls

Security Area	Control
Authentication	Secure password hashing, session expiry, login attempt monitoring, password reset tokens and optional 2FA readiness.
Authorization	Database-backed RBAC, station scoping, server-side permission checks and restricted administrative actions.
Sensitive Data	Field masking and permission restriction for salary, ID, wallet and other sensitive records; encryption applied where appropriate.
Transport Security	HTTPS/TLS enforced for all production traffic.
File Security	Private S3 bucket, presigned URLs, file validation and no direct public object exposure.
Input Security	Zod/server validation, safe database access, upload restrictions and output encoding.
Financial Integrity	Verified transactions are reversed rather than silently edited; approvals and reasons required for sensitive corrections.
Audit Trail	Immutable logs for login, user changes, stock changes, price changes, sales, refunds, reversals, approvals and management utilities.
Session Management	Account disabling, role change enforcement, session timeout and active-session revocation capability.
Operational Security	Separate environments, least-privilege cloud credentials, secret rotation and restricted production access.
Backup & Recovery	Automated backups, restore procedures and periodic recovery validation.
Compliance Readiness	Architecture supports future formal policies for data retention, privacy notices, incident response and access reviews.

10. Project Deliverables

Category	Deliverable
Application	Production-ready responsive Next.js web application covering the agreed modules.
Source Code	Organized TypeScript source repository with environment templates and deployment configuration.
Database	Prisma schema, migration history, seed data and production database configuration.
Cloud Deployment	Configured Vercel projects/environments, custom domain connection and deployment workflow.
File Storage	Configured private AWS S3 bucket integration with signed upload/download access.
Authentication & Security	RBAC, account lifecycle controls, session handling, audit trail and protected sensitive operations.
Documents & Printing	Receipt, invoice, cargo label, statement and report templates for thermal/A5 printing.
Reporting	Role-aware dashboards, filters and exports for operational and financial reporting.
Documentation	Installation/deployment guide, environment-variable guide, user/admin guide and backup/recovery notes.
Training	Administrative and operational training sessions for nominated users.
Testing	Core workflow testing, UAT support and correction of verified defects before final acceptance.
Handover	Production credentials ownership transfer, source access, documentation and signed acceptance record.

11. Acceptance Criteria

Area	Acceptance Standard
Authentication & RBAC	Approved users can log in and can access only authorized modules and station scopes.
Station Operations	Management can create/configure stations, assign users and view station-specific and consolidated information.
Inventory	Stock-in, stock-out, transfer, adjustment, damage and return transactions update traceable inventory balances.
POS & Payments	Authorized users can create sales, accept configured payment methods and generate numbered receipts/invoices.
Cargo Labels	Users can create, search and reprint AWB/cargo labels containing required fields, barcode and QR code.
Agents & Wallets	Deposits, debits, credits, balances and statements reconcile against an immutable transaction history.
Finance & Reports	Income, expenses, refunds, station revenue, profit and filtered exports produce consistent results from recorded data.
Audit	Sensitive actions record actor, time, station, action and related record; correction activity is traceable.
Files	Authorized users can upload and retrieve documents through protected S3 access.
Deployment	Staging and production environments are deployed, secured by HTTPS and connected to the approved domain.
Responsive UI	Core workflows are usable on common desktop, tablet and mobile screen sizes.
UAT	Client-nominated users execute and sign off the agreed UAT checklist or provide documented defects for correction.

Final Agreement Notes: The following items should be confirmed in writing before development begins or before the relevant implementation phase. These notes are intended to prevent ambiguity, protect both parties and keep delivery aligned with the six-week plan.

Item to Agree	Recommended Position / Decision Required
Final Scope Baseline	This proposal and its module table become the baseline. Any material addition, new workflow or third-party integration is handled through a written change request.
Commercial Fee	Insert and approve the total project fee, applicable taxes and any reimbursable expenses.
Payment Milestones	Recommended structure: 40% mobilization, 30% after Week 3 production release, 20% after UAT delivery, and 10% after final acceptance/handover.
Cloud Costs	Vercel, AWS, managed PostgreSQL, domain, email/SMS, monitoring and third-party services are paid directly by the client or reimbursed separately from development fees.
Cloud Account Ownership	Production Vercel, AWS, database and domain accounts should be owned by AAU Chamo, with controlled technical access granted to Nillar.
Source Code Ownership	Upon full payment, AAU Chamo receives the project-specific source code and deployment materials. Reusable frameworks, generic utilities and pre-existing Nillar intellectual property remain with Nillar unless otherwise agreed.
Confidentiality	Both parties protect credentials, customer records, financial information, employee information and proprietary business processes.
Data Migration	Manual entry and agreed import templates are included. Cleaning or migrating large

	legacy datasets is separately scoped after inspection.
Offline Mode	Offline capability will prioritize designated operational transactions through local encrypted queues and conflict-aware synchronization. Full offline parity for every module is not assumed unless separately specified.
Hardware	Thermal printers, A5 printers, barcode scanners, computers, network equipment and consumables are supplied/configured by the client; compatibility will be validated using provided models.
Airline Integration	Manual flight booking records are included. Live airline/GDS booking, fare search or ticket issuance requires separate provider contracts, credentials, fees and integration scope.
Accounting/Payment Integration	Internal financial records and payment-method capture are included. Live bank, POS, accounting or payment-gateway integrations require separate API access and approval.
UAT Window	AAU Chamo should provide consolidated UAT feedback within 3 business days of each scheduled review. Delayed feedback may shift milestones.
Defect Warranty	Recommended: 30 calendar days after final acceptance for correction of reproducible defects within approved scope. Enhancements are not defects.
Support & Maintenance	After the warranty period, support may continue under a monthly or annual maintenance agreement covering monitoring, minor fixes, updates, backups and agreed service levels.
Availability Target	Cloud services are dependent on third-party providers and internet connectivity. A formal uptime SLA requires an ongoing support plan and appropriate paid hosting tiers.
Backup Retention	Confirm daily backup retention period, restore test frequency and whether S3 cross-region replication is required.
Security Governance	AAU Chamo will nominate a data owner and security administrator, review active users periodically and report suspected compromise immediately.
Change Control	Any approved change records description, impact, cost, timeline and acceptance effect before implementation.
Termination/Handover	On termination and settlement of outstanding obligations, project materials and client-owned credentials/data will be handed over in a reasonable export format.
Proposal Validity	Commercial and delivery assumptions remain valid for 30 days and may be revised if provider prices or scope materially change.

14. Assumptions and Exclusions

Classification	Detail
Assumption	The client will provide accurate business rules, tax settings, pricing, user roles, stations, products, agents and opening balances.
Assumption	A stable internet connection is available at stations for full real-time functionality; offline support is a resilience feature, not a substitute for connectivity.
Assumption	The client has authority to collect and process customer, staff, passenger and financial information entered into the system.
Exclusion	Purchase of computers, printers, scanners, network devices, SIM/data plans and other physical hardware.
Exclusion	Third-party subscription, transaction, messaging, storage, domain, airline/GDS, payment or accounting fees.
Exclusion	Full payroll, statutory tax filing, complete accounting general ledger or banking integration unless separately approved.
Exclusion	Native Android/iOS mobile applications; the included interface is responsive and may be delivered as a PWA.

Exclusion	Unlimited historical-data migration, data cleansing or physical document digitization.
Exclusion	Features not listed in the final scope table or approved through change control.